QP Code: II SEMESTER M.TECH DEGREE EXAMINATION, XXX 20XX

Total Pages: 3

Reg No.:_______________ **Name:**______________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Course Code: 222ECS002

Course Name: DEEP LEARNING

Max. Marks: 60 **Duration: 2.5 Hours**

Limit answers to the required points.

PART A

Answer all questions, each carries 5 marks. Marks

1 Explain how Lasso regression method leads to weight sparsity. (5)

2 Weight sharing allows CNNs to deal with image data without using too many parameters. Does weight sharing increase the bias or the variance of a model? Explain. (5)

3 Show the steps involved in an LSTM to predict stock prices. Give one advantage of using an RNN rather than a convolutional network. (5)

4 In what ways is a Markov Chain Monte Carlo (MCMC) approach different from other types of Monte Carlo simulations? (5)

5 "An auto encoder is a neural network that learns to copy its input to its output." Justify the above statement with the help of the structural diagram of the auto encoder. (5)

PART B

Answer any FIVE full question, each question carries 7 marks.

6 a) Discuss the advantages of the online version of Gradient Descent as compared to normal Gradient Descent.Write the equation for weight updation in online Gradient Descent. (7)

7 a) Consider an input of size [illegible]13 and a filter of size [illegible]. Calculate the size of the [illegible] map after performing [illegible] with strides 1 and 2. (4)

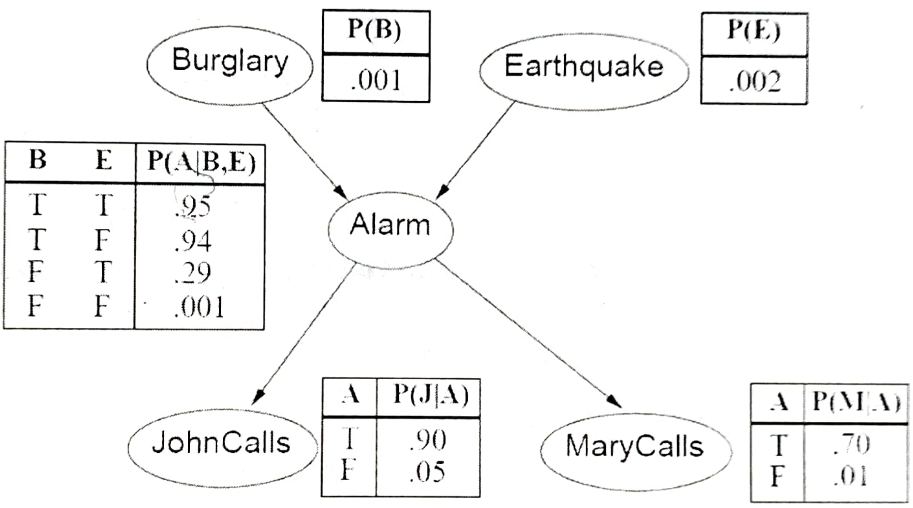
b) Discuss the motivations for pooling in neural networks. (3)

8 a) The vanishing gradient problem is more pronounced in RNN than in traditional neural networks. Give a reason. Discuss a solution for the problem. (4)

b) List the differences between LSTM and GRU (3)

9 a) Shown below is the Bayesian network corresponding to the Burglar Alarm problem, P(J | A) P(M | A) P(A | B, E) P(B) P(E). The probability tables show the probability that variable is True, e.g., P(M) means P(M = t). (7)

Calculate **P(J = t ∧ M = f ∧ A = f ∧ B = f ∧ E = t)**.



P(B)
.001

P(E)
.002

B	E	P(A|B,E)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

10 a) Based on the formula for the energy function (E) of a Restricted Boltzmann Machine (RBM), list the difference between Boltzmann Machine and Deep Belief Network. (5)

b) Generative Adversarial Networks(GANs) include a generator and a discriminator. Sketch a basic GAN using those elements, a source of real images, and a source of randomness. (2)

11 a) Initializing the weights of a neural network with very small or large random numbers is not advisable. Justify. (3)

b) Weight sharing allows CNNs to deal with image data without using too many parameters. Does weight sharing increase the bias or the variance of a model? Explain (4)

12 a) Compare Boltzmann Machine with Deep Belief Network (7)

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